



INDEPENDENT STUDIES

FOURTH SESSION

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3,130 Corporate Finance

Major in Business Administration (BWL)

9. Conflicts of Interest and Corporate Governance (BMAE, Chapter 19)

9.1. True / False Questions

- a. The separation of ownership and control can result in managerial shirking.
 - a. **True**
 - b. False
- b. The principal-agent model describes the separation of ownership and control.
 - a. **True**
 - b. False
- c. An independent board of directors, large shareholders, and competing management teams / companies are effective monitoring mechanisms of company management.
 - a. **True**
 - b. False
- d. More prohibitive takeover laws lead to better corporate governance.
 - a. True
 - b. **False**
- e. When facing a debt overhang problem, firms with existing senior debt obligations may not be able to finance positive NPV investments.
 - a. **True**
 - b. False
- f. Managerial empire building should be incentivized by shareholders because it leads to larger and better performing firms.
 - a. True
 - b. **False**
- g. Based on the majority of empirical research on the topic, good corporate governance increases the expected return on equity to the extent that it reduces shareholders' monitoring costs.
 - a. True
 - b. **False**
- h. A large fraction of compensation for U.S. CEOs is made up by grants of restricted shares or performance shares.
 - a. **True**
 - b. False

- i. Stock-option grants give the manager a certain number of shares delivered at annual intervals, usually over five years.
 - a. True
 - b. False**

9.2. Risk shifting

Assume that a firm has two mutually exclusive investment projects A and B to choose from with the following characteristics:

	Investment	Payoff bad state ($p=0.5$)	Payoff good state ($p=0.5$)
Project A	175	125	250
Project B	175	50	300

Assume that the management can raise the necessary funds by either issuing equity or by issuing a combination of equity and unprotected debt of 90. How much do equity holders gain, if the project is financed with a combination of debt and equity rather than equity only? For simplicity, you can assume that all actors are risk-neutral and that the risk-free rate is zero.

- a. 2.00
- b. 5.00
- c. 7.50**
- d. 12.50
- e. 20.00

Financed with equity only:

$$\text{Value of equity}(A) = -175 + 125 \cdot 0.5 + 250 \cdot 0.5 = 12.5$$

$$\text{Value of equity}(B) = -175 + 50 \cdot 0.5 + 300 \cdot 0.5 = 0$$

→ Under full equity-financing, shareholders prefer Project A (and gain 12.5)

Financed with combination of debt of 90 and equity of 85:

$$\text{Value of equity}(A) = -(175 - 90) + (125 - 90) \cdot 0.5 + (250 - 90) \cdot 0.5 = 12.5$$

$$\text{Value of equity}(B) = -(175 - 90) + (300 - 90) \cdot 0.5 = 20$$

→ Under combined debt and equity financing, shareholders prefer Project B

→ Debt holders bear downside risk; equity is wiped out in bad state in Project B ($50 < 90$)

$$\text{Gain by equity holders from debt financing} = 20 - 12.5 = 7.5$$

9.3. Bait and switch

A firm has debt of 125 outstanding. There are two equally likely firm values next year. In the good state, the firm will be worth 200. In the bad state, the firm will be worth 140. How much will bondholders lose, if the firm raises new debt of 50 today, with the same seniority as the existing debt and immediately pays 50 as dividend to shareholders?

- a. 5.25
- b. 8.75
- c. 12.50
- d. 17.50**
- e. 22.25

Before the debt issue, debt is safe in both states as the company can repay the debt in the good state ($200 > 125$) and in the bad state ($140 > 125$):

$$D_{\text{before refinancing}} = 0.5 \cdot 125 + 0.5 \cdot 125 = 125$$

After raising new debt, the company is unable to repay the face value in the bad state and defaults ($140 < (125 + 50)$):

$$D_{\text{after refinancing}} = 0.5 \cdot 175 + 0.5 \cdot 140 = 157.5$$

⇒ The bondholders lose 17.50 ($= 175 - 157.5$)

9.4. Debt overhang problem

Which of the following statements describes the debt overhang problem best?

- a. **Firms with existing senior debt obligations may not be able to finance positive NPV investments because the payoff from the new investments goes in large parts (or even completely) to the existing bondholders due to their seniority.**
- b. Firms with existing junior debt obligations may not be able to finance positive NPV investments because the payoff from the new investments goes completely to the new senior bondholders due to their seniority.
- c. Firms with existing senior debt obligations may only be able to finance negative NPV investments if new senior bondholders are willing to gamble.
- d. Firms with existing senior debt obligations may not be able to finance any positive NPV investments because the payoff from the new investments goes completely to the newly raised junior debt due to their seniority.
- e. None of the answers a. to d. are correct.

9.5. Free rider problem

A firm has assets in place that will generate USD 252 next year, at which point the company will be liquidated. The firm is financed with equity and four senior bonds with face value of USD 105 each. The firm cannot issue debt with a higher priority than the four outstanding senior bonds. Now a new investment project comes up that has to be decided on today. It requires an initial investment of USD 28 and will pay out USD 84 next year.

Assume that one debt holder steps in and buys new debt of USD 28 with the same priority as the existing debt, while the other three debt holders decline to further invest in this firm. How much does each non-participating debt holder gain from the new investment project?

- a. USD 10.75
- b. USD 11.50
- c. USD 13.25
- d. USD 14.75
- e. **USD 15.75**

Payoff with no investment:

$$(252 / 4) - 105 = -42$$

Payoff with investment:

$$\text{Non-Participant: } 105 / (420 + 28) * (252 + 84) - 105 = -26.25$$

$$\text{Difference } 42 - 26.25 = 15.75$$

9.6. Economic Value Added (EVA)

One potential metric used to measure and reward top managers for performance is EVA. Omega Corporation is a small but profitable producer of dietary supplements for pets. This is not a high-tech business, but Omega's earnings have averaged around USD 1.2 million after tax, largely on the strength of its patented enzyme for making cats nonallergenic. The patent has eight years to run, and Omega has been offered USD 4 million for the patent rights. Omega's assets include USD 2 million of working capital and USD 8 million of property, plant, and equipment. The patent's value does not show in Omega's books. Suppose Omega's cost of capital is 15%.

What is its EVA?

- a. USD −1.05 million
- b. USD −0.9 million**
- c. USD 0 million
- d. USD 0.6 million
- e. USD 0.9 million

$$\begin{aligned} EVA &= \text{residual income} = \text{income earned} - \text{income required} \\ &= \text{income earned} - \text{cost of capital} \cdot \text{investment} \end{aligned}$$

$$EVA = 1.2 - 0.15 \cdot (4 + 2 + 8) = -0.9$$

9.7. Interpreting Regression Tables

The following table reports results of regressions of Tobin's Q on a corporate governance index, control variables, and different fixed effects.

The table reports estimates from an OLS regression (Column 1) and a fixed effects regression (Column 2) of Tobin's Q on a corporate governance index and control variables. Both regressions include year fixed effects. Column 1 additionally includes industry fixed effects. Column 2 additionally includes firm fixed effects (but no industry fixed effects). The sample includes 11,222 observations on 1,912 listed U.S. firms from 2012 to 2017. t-statistics are in parentheses. ***, **, * denotes statistical significance at the 1%, 5%, 10% level.

Dependent Variable:	Tobin's Q	
	(1)	(2)
Corporate Governance Index	0.204 *** (2.89)	0.172 ** (2.06)
Firm Size	-0.023 (1.01)	-0.017 (0.89)
Leverage	-0.245 ** (2.04)	0.123 (1.17)
R&D Expenditures / Sales	0.784 * (1.91)	0.813 ** (2.11)
Capital Expenditures / Sales	0.306 * (1.87)	0.287 (0.94)
Year Dummies	Yes	Yes
Industry Dummies	Yes	No
Firm Fixed Effects	No	Yes
Observations	11,222	11,222
R-Squared	0.32	0.23

Which of the following statements is not correct?

- The regression table shows a positive and statistically significant relationship between a corporate governance index and Tobin's Q.
- The OLS specification in Column 1 controls for both general time trends and differences in Tobin's Q across industries.
- The firm fixed effects regression in Column 2 transforms the cross-sectional regression in Column 1 into a within-firm analysis, showing how Tobin's Q changes on average in response to a change in the governance index.
- Leverage is negatively and significantly related to Tobin's Q in the OLS regression in Column 1, but insignificantly related to Tobin's Q in the fixed effects regression in Column 2. Hence, in the cross-section, different levels of leverage are associated with different levels of firm value. In contrast, changes in leverage are not significantly related to (contemporaneous) changes in Tobin's Q.
- Capital expenditures over sales are positively and significantly related to Tobin's Q in the OLS regression in Column 1, but insignificantly related to Tobin's Q in the fixed effects regression in Column 2. Hence, a change in capital expenditures is associated with a significant change in Tobin's Q, while different levels of capital expenditures are not associated with different levels of firm value.**

10. Market Efficiency and Behavioral (Corporate) Finance (BMAE, Chapter 12)

10.1. True / False Questions

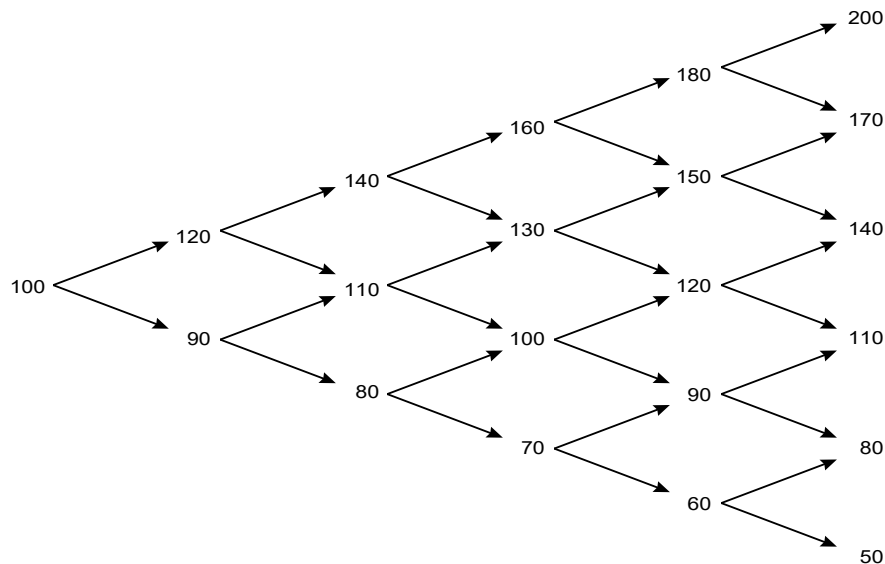
- a. The random increments of a random walk are independent and identically distributed. Therefore, they must have a positive serial correlation.
 - a. True
 - b. False**
- b. The standard deviation of a random variable modeled with a random walk increases exponentially in t .
 - a. True
 - b. False**
- c. The expected value of a random variable modeled with random walk with a positive drift increases linearly in t .
 - a. True**
 - b. False
- d. In the semi-strong form of market efficiency, private information are reflected in the prices.
 - a. True
 - b. False**
- e. If the price of an asset is equal to its fair value, there are no arbitrage opportunities.
 - a. True**
 - b. False
- f. Behavioral finance tries to explain why prices can depart from their fundamental values.
 - a. True**
 - b. False
- g. Kahneman and Tversky provided experimental evidence that supports the hypothesis that investors are risk-averse regarding choices involving losses but risk-seeking regarding choices involving gains.
 - a. True
 - b. False**
- h. The inconsistent attitude of investors towards gains and losses is often subsumed under the “disposition effect”.
 - a. True**
 - b. False
- i. In the context of investments, overconfidence describes the behavior of market participants to assign overly large confidence intervals to their estimates.

- a. True
 - b. False**
 - j. In the context of investments, the anchoring bias describes the tendency of investors to overly rely on the first information provided.
 - a. True**
 - b. False
 - k. If a random variable has a positive drift, its returns cannot be stationary.
 - a. True
 - b. False**
 - l. If a random walk is i.i.d. (independent, identically distributed) the probability of a positive return is larger after a negative return than after a positive return.
 - a. True
 - b. False**
 - m. In an efficient market, information is costless.
 - a. True**
 - b. False
 - n. If markets are efficient, shareholders should expect to receive only the risk-free interest rate on their investment.
 - a. True
 - b. False**
 - o. The small-firm effect provides evidence against market efficiency.
 - a. True**
 - b. False
 - p. If there are no arbitrage opportunities, prices are always equal to their fair value.
 - a. True
 - b. False**
- not necessary: e.g., if you cannot trade, prices can differ although there are no arbitrage opportunities.
- q. Consider the example of eBay. The only reason that could lead an analyst to be over-optimistic about the company's forecasts of earnings and cash flows is an agency conflict the analyst may face.
 - a. True
 - b. False**

→ Overoptimism can also come from the overconfident belief that a company will perform well e.g., eBay valuation example from the lecture.

10.2. Random Walk

A share price follows a random walk. It is modeled by a binomial tree with the starting price of 100. That is the price in t is 100, in $t+1$ it is 120 or 90, in $t+2$... etc. We assume that the probabilities for an up-move (e.g. from 100 to 120) and a down-move (e.g. from 100 to 90) are the same.



10.2.1. What is the expected value and standard deviation of the price at $t+1$?

- a. Expected value is 105 and standard deviation is 225
- b. Expected value is 105 and standard deviation is 15**
- c. Expected value is 100 and standard deviation is 15
- d. Expected value is 110 and standard deviation is 225
- e. None of the answers a. to d. are correct.

$$E(value)_1 = \frac{1}{2} \cdot 120 + \frac{1}{2} \cdot 90 = 105$$

$$Std = \sqrt{\frac{1}{2} \cdot (120 - 105)^2 + \frac{1}{2} \cdot (90 - 105)^2} = 15$$

10.2.2. What is the expected value and standard deviation of the price at t+2?

- a. **Expected value is 110 and standard deviation is 21.21**
- b. Expected value is 105 and standard deviation is 18.12
- c. Expected value is 100 and standard deviation is 450.00
- d. Expected value is 110 and standard deviation is 30.00
- e. None of the answers a. to d. are correct.

$$E(value)_2 = \frac{1}{2} \cdot \frac{1}{2} \cdot 140 + \frac{1}{2} \cdot \frac{1}{2} \cdot 110 + \frac{1}{2} \cdot \frac{1}{2} \cdot 110 + \frac{1}{2} \cdot \frac{1}{2} \cdot 80 = 110$$

$$Std = \sqrt{\frac{1}{4} \cdot (140 - 110)^2 + \frac{1}{4} \cdot (110 - 110)^2 + \frac{1}{4} \cdot (110 - 110)^2 + \frac{1}{4} \cdot (80 - 110)^2} = 21.21$$

10.2.3.

The increments of a random walk of a two-point distribution have the realizations of +15 with probability 0.7 and -5 with probability 0.3.

What is the drift of the random walk?

- a. 10
- b. 11
- c. 12
- d. **9**
- e. None of the answers a. to d. are correct.

$$E(drift) = 0.7 \cdot 15 + 0.3 \cdot (-5) = 9$$

10.2.4.

What is the variance of the random variable described in exercise 8.2.4?

- a. 0.458.
- b. 4.200.
- c. 4.583.
- d. 9.165.
- e. **None of the answers a. to d. are correct.**

$$Var = 0.7 \cdot (15 - 9)^2 + 0.3 \cdot (-5 - 9)^2 = 84$$

10.2.5.

What is the expected value of the random variable described in exercise 8.2.4 after 5 periods?

- a. $X_0 + 20$.
- b. **$X_0 + 45$.**
- c. $X_0 + 50$.
- d. $X_0 + 108$.
- e. None of the answers a. to d. are correct.

$$E(X) = X_0 + 5 \cdot 9 = X_0 + 45$$

10.2.6.

What is the variance of the random variable described in exercise 8.2.4 after 5 periods?

- a. 10.
- b. 20.
- c. 30.
- d. 420.**
- e. None of the answers a. to d. are correct.

$$Var = 5 \cdot 84 = 420$$

10.2.7.

The statement that stock prices follow a random walk implies that:

- I) successive price changes are independent of each other;
- II) successive price changes are positively related;
- III) successive price changes are negatively related;
- IV) the autocorrelation coefficient is either +1.0 or -1.0

- a. I only**
- b. II and III only
- c. IV only
- d. III only
- e. None of the answers a. to d. are correct.

10.3. Yield on stock

Assume that continuous returns are normally distributed and that the return of your portfolio is distributed as $\sim N(0.05, 0.1)$. (Hint: $r \sim N(\mu, \sigma)$)

10.3.1.

What is the most likely probability that your portfolio will get a negative return?

- a. 30.9%**
- b. 48.0%
- c. 50.0%
- d. 69.1%
- e. 83.2%

$$p = N\left(\frac{0 - 0.05}{0.1}\right) = N(-0.5) = 0.309$$

10.3.2.

What is the most likely probability that your portfolio will get at least a return of 15%?

- a. 10.3%
- b. 15.9%**
- c. 44.0%
- d. 56.0%
- e. 84.1%

$$p = 1 - N\left(\frac{0.15 - 0.05}{0.1}\right) = 1 - N(1) = 1 - 0.841 = 0.159$$

10.4. Efficient Markets

10.4.1.

Under the assumption that capital markets are efficient, the sale or purchase of any security at the prevailing market price is generally...

- a. ...a positive-NPV transaction.
- b. ...a zero-NPV transaction.**
- c. ...a negative-NPV transaction.
- d. ...no general trend exists for such transactions.
- e. None of the answers a. to d. are correct.

10.4.2.

Which of the following statements is/are true if the strong-form efficient market hypothesis holds?

- I) Analysts can easily forecast stock price changes.
- II) Financial markets are irrational.
- III) Stock returns follow a particular pattern.
- IV) Stock prices reflect all available information.

- a. I only
- b. II only
- c. I and III only
- d. IV only**
- e. None of the answers a. to d. are correct.

10.4.3.

Predictable cycles in stock price movements:

- I) tend to persist for a long time;
- II) tend to self-destruct as soon as investors recognize them;
- III) never appear, since stock returns change randomly

- a. I only
- b. II only**
- c. III only
- d. I, II, and III
- e. None of the answers a. to d. are correct.

10.5. Abnormal returns

An analysis of 60 monthly returns of company Epsilon's common stock provides a beta of 1.45 and the monthly risk-free rate is 0.2%. One month later, the market is up by 5% over the risk-free rate, and Epsilon's stock is up by 6%.

What is Epsilon's abnormal rate of return?

- a. - 5.55%
- b. - 3.15%
- c. - 1.45%**
- d. - 0.2%
- e. 7.05%

$$\text{Expected return} = 0.002 + 1.45 \cdot 0.05 = 0.0745 = 7.45\%$$

$$\text{Abnormal return} = 0.06 - 0.0745 = -0.0145 = -1.45\%$$

10.6. Behavioral Finance

Mr. Warren Gates is a rational investor with a concave utility function for money $u(\text{money})$. His utility for CHF 0 is zero, i.e. $u(0) = 0$. On the market he is offered two lotteries:

Lottery I:

(a) Certain amount of CHF 1.000

(b) Lottery that pays as follows: CHF 5.000 with 10% chance;
CHF 1.000 with 89% chance;
CHF 0 with 1% chance.

Lottery II:

(a) CHF 1.000 with 11% chance or CHF 0 with 89% chance.

(b) CHF 5.000 with 10% chance or CHF 0 with 90% chance.

Mr. Warren Gates looks at Lottery I and decides for option (a). Which option will he choose for Lottery II?

- a. He will choose Lottery II (a).**
- b. He will choose Lottery II (b).
- c. He is indifferent between both lotteries.
- d. We cannot say without having a defined utility function $u(x)$.
- e. None of the answers a) to d) are correct.

Since in Lottery I (a) is preferred to (b), we have:

$$1 \cdot u(1000) > 0.1 \cdot u(5000) + 0.89 \cdot u(1000) + 0.01 \cdot u(0) \rightarrow 0.11 \cdot u(1000) > 0.1 \cdot u(5000)$$

If Lottery II (b) is preferred to Lottery II (a), then:

$$0.11 \cdot u(1000) < 0.10 \cdot u(5000)$$

From the first inequality (Lottery I), we know that the investor prefers (a): $0.11u(1000) > 0.10u(5000)$. This would be a contradiction to his second choice. Therefore, Mr. Warren Gates as a rational investor will choose Lottery II (a)

10.7. A behavioral bias: Overconfidence

Consider the example “eBay’s valuation”. Which statement regarding Mary Meeker’s valuation of eBay is correct?

- a. Mary Meeker was the only analyst who was overconfident about internet companies’ outstanding growth prospects.
- b. Mary Meeker intentionally inflated eBay’s revenue forecast in order to display the highest possible valuation of the company.
- c. The analysts following eBay were so pessimistic about eBay’s future revenue stream that their value per share of the company was significantly different than Meeker’s target price per share.
- d. Each of the three valuation techniques used by Meeker gave the same target price per share for eBay.
- e. **Meeker’s belief was overconfident in the ability of eBay to become the main actor of the “wal-marting of the web” and then dominate internet commerce.**

11. Empirical Research in Finance

11.1. True / False Questions

- a. Tobin's Q is a widely used proxy in empirical corporate finance to measure firm performance.
 - a. **True**
 - b. False

- b. If Tobin's Q is lower than 1.0, the market value of the firm is greater than the value of the company's recorded assets.
 - a. True
 - b. **False**

- c. The G-Index comprises 24 antitakeover provisions and is widely used as a measure of corporate governance.
 - a. **True**
 - b. False

- d. When investigating the effect of corporate governance on firm value, one of the issues is that most data is available in yearly intervals only.
 - a. **True**
 - b. False

- e. The t-statistic is defined as the coefficient estimate multiplied by the standard deviation of the estimate.
 - a. True
 - b. **False**

- f. In a multivariate regression setting, the coefficient estimate of one variable shows the effect of this particular variable on the dependent variable while holding all other variables in the regression constant.
 - a. **True**
 - b. False

- g. Adding industry dummy variables to a regression controls for differences in the dependent variable across industries.
 - a. **True**
 - b. False

- h. "Natural experiments" are one possible way to mitigate endogeneity concerns.
 - a. **True**
 - b. False

11.2. The effect of currency derivatives on Tobin's Q

This table presents results of a pooled OLS regression of the natural logarithm of Tobin's Q on a "FCD dummy" and different control variables. The FCD dummy equals 1 if a firm uses foreign currency derivatives, and 0 otherwise. The table reports coefficient estimates and t-statistics. *, **, *** indicate statistical significance at the 10%, 5%, 1% level, respectively.

Dependent variable: $\ln(\text{Tobin's } Q)$	(1)
Observations	2069
R^2	0.73
FCD dummy (% of Q in parenthesis)	0.053 (5.26%)
	2.989***
Foreign sales/total sales	0.163
	4.229***
Size (log of total assets)	-0.071
	-7.790***
ROA	0.030
	11.335***
Debt to equity	0.000
	5.004***
Growth (capital exp/sales)	0.131
	1.367
Diversification dummy	-0.102
	-4.830***
Dividend dummy	-0.090
	-3.803***
Advertising/assets	1.173
	3.879***
R&D/assets	-0.840
	-2.330**

Which statement regarding the table is correct?

- Tobin's Q is calculated as the ratio of the market value of the firm and replacement cost of assets.**
- For a one standard deviation increase in FCD, the natural logarithm of Tobin's Q increases by 5.26%.
- The coefficient of the FCD dummy is positive but insignificant so that there is no reliable relationship between using currency derivatives and Tobin's Q.
- Firms which use currency derivatives have a Tobin's Q that is lower by 5.3% than that of nonusers.
- None of the answers a) to d) are correct.

11.3. Interpreting Regression Tables

The following tables present summary statistics and results of the regression of cumulative abnormal returns (CAR) on different explanatory variables.

Summary Statistics

This table reports summary statistics of firm-level variables. The sample period is 2001–2007. *Market cap* (market capitalization) is the product of stock price at calendar year-end (item *prcc_c*) and shares outstanding in millions (item *csho*). $\ln()$ is the natural logarithm. *Book leverage* is total liabilities (item *lt*) scaled by total assets (item *at*). *Liquidity* is cash and short-term investments (item *che*) scaled by total assets (item *at*). The *credit rating* is the median domestic long-term issuer credit rating in a given year. Letter ratings are transformed into numerical ones as follows: 9="AAA," 8="AA," 7="A," 6="BBB," 5="BB," 4="B," 3="CCC," 2="CC," 1="C," and 0="D." All variables (except *credit rating*) are winsorized at the 1st and 99th percentiles. SD is the standard deviation, P25 the first and P75 the third quartile of the respective variable. N indicates the number of observations.

	Mean	Median	SD	P25	P75	N
<i>Market cap</i>	26,080	6,860	51,655	1,499	23,506	1,413
<i>ln(Market cap)</i>	8.67	8.83	1.96	7.31	10.07	1,413
<i>Book leverage</i>	0.62	0.64	0.23	0.47	0.8	1,413
<i>Liquidity</i>	0.14	0.08	0.17	0.03	0.19	1,413
<i>Credit rating</i>	6.27	6	1.23	6	7	1,413

Regression Table

In this table, 21-day cumulative abnormal returns (CAR) are regressed on several firm-level characteristics. $\ln()$ is the natural logarithm. *Liquidity* is cash and short-term investments scaled by total assets. *Book leverage* is defined as total liabilities scaled by total assets. The *credit rating* is the (numerical) median domestic long-term issuer credit rating in the year in which an event occurs. CARs are calculated with respect using a market model. The *t*-statistics (in parentheses) are based on robust standard errors. (* $p < 0.10$, ** $p < 0.05$ *** $p < 0.01$)

Dependent variable	(1) CAR	(2) CAR
<i>Liquidity</i>		-0.102*** (-3.56)
<i>Book leverage</i>	0.032** (2.01)	
<i>Credit rating</i>	0.006 (1.37)	0.004 (0.84)
<i>ln(Market cap)</i>	-0.005 (-1.54)	-0.003 (-1.01)
<i>Constant</i>	-0.015 (-0.59)	0.017 (0.77)
Observations	1,413	1,413
R ²	0.015	0.032

Which of the following statements is false?

- a. The regression table shows only a statistically significant relationship between *Book leverage* and cumulative abnormal returns and between *Liquidity* and cumulative abnormal returns.
- b. A one-standard deviation increase in *Book leverage* increases cumulative abnormal returns by 0.736%.
- c. A one-standard deviation decrease in *Liquidity* increases cumulative abnormal returns by 1.734%.
- d. A one-standard deviation increase in *Liquidity* is associated with an average decrease in firms' market capitalization by approximately 452.23.
- e. **A one-standard deviation increase in *Market cap* in column (1) is associated with an average decrease in firms' market capitalization by approximately 255.58.**

Statement a. is true.

Statement b. is true: $0.23 \cdot 0.032 = 0.00736$

Statement c. is true: $-0.17 \cdot (-0.102) = 0.01734$

Statement d. is true: $0.17 \cdot (-0.102) \cdot 26,080 = -452.23$

Statement e. is false. To be true, it would need say: "A one-standard deviation increase in *Ln(Market cap)* in column (1) is associated with an average decrease in firms' market capitalization by approximately 255.58."

$$1.96 \cdot (-0.005) \cdot 26,080 = -255.58$$

11.4. Interpreting Regression Tables

The following table reports results of regressions of Tobin's Q on CEO tenure and various control variables.

Regression Table: The relation between CEO tenure and firm value

This table reports results from firm fixed effects regressions of Tobin's Q on different functional specifications of CEO tenure along with CEO, firm and corporate governance characteristics. The sample includes S&P 1500 firms over the period 1998-2011 excluding regulated utility and financial firms (SIC codes 4000-4999 and 6000-6999). Tobin's Q is calculated as total assets minus book equity plus market value of equity all divided by total assets. CEO tenure is the number of years the CEO has been serving as the firm's CEO. CEO characteristics are defined as follows: CEO age is the age of the CEO in years. CEO gender is a dummy equal to one if the CEO is female. CEO power index is the sum of the following indicator variables: CEO ownership above median, Co-Option above median, Duality, Involved CEO, President, Only insider. The index can take on values between zero and six. Founder CEO is a dummy equal to one if the CEO is a founder of the company. Specification (3) shows results of re-estimating regression specification (2) with interacted year and industry (based on Fama French 48 industries) fixed effects as additional controls. An intercept and year dummies are included in all regressions, but not reported. Turning point refers to the inflection point of CEO tenure in the non-linear specifications. Robust t-statistics are reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

Dependent Variable: Tobin's Q			
	(1)	(2)	(3)
CEO tenure	-0.0021 (-0.483)	0.0146** (2.410)	0.0130** (2.162)
CEO tenure squared		-0.0006*** (-3.619)	-0.0006*** (-3.701)
<i>CEO characteristics</i>			
CEO age	0.0024 (0.828)	0.0019 (0.652)	0.0040 (1.393)
CEO gender	0.0318 (0.365)	0.0411 (0.476)	0.0131 (0.163)
CEO power index	-0.0080 (-0.425)	-0.0225 (-1.162)	-0.0252 (-1.302)
Founder CEO	0.3802** (2.086)	0.4236** (2.283)	0.3783** (2.073)
Corporate governance characteristics	Yes	Yes	Yes
Firm characteristics	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Industry x year fixed effects	No	No	Yes
Observations	12,427	12,427	12,427
R-squared (within)	0.295	0.297	0.354
Turning point (yrs)	-	12.2	10.8

Which of the following statements is false?

- The regression table shows a concave relationship between CEO tenure and firm value: Starting at the appointment of a CEO, firm value on average increases up to a CEO tenure of 11-12 years and then declines.
- Neglecting the non-linearity in the relationship between CEO tenure and firm value, we would fail to find a significant relationship.
- CEOs who are founders are on average associated with higher firm-values.

- d. Controlling for shocks at the industry-level by using industry x year fixed effects in column (3) significantly changes the results reported in column (2).**
- e. The coefficients on CEO age are statistically insignificant. Hence, there is no evidence of a statistically meaningful relationship between CEO age and firm value.

Note on a.: the inflection point x of a quadratic function, $ax^2 + bx + c$, can be calculated as follows: $-\frac{b}{2a} = -\frac{0.0146}{2(-0.0006)} = 12.1667$.